📊 Flowchart Sistem Informasi Pegawai Honorer

Dokumen ini berisi **dua flowchart terpisah** hasil analisis mendalam terhadap seluruh migration, model, controller, route, dan data JSON ekspor database.

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1. Flowchart Database (Entity Relationship Diagram)

Diagram di bawah menunjukkan **struktur tabel, kolom, tipe data, dan relasi antar tabel** beserta constraint foreign key.

```
erDiagram
    satkers {
        bigint id PK
        varchar nama_satker UK "unique"
        timestamp created_at
        timestamp updated_at
    }

    users {
        bigint id PK
        varchar name
        varchar username UK "unique, renamed from email"
        varchar password
        varchar remember_token
        enum role "super_admin | admin_satker"
        bigint satker_id FK "nullable, nullOnDelete"
        enum status "active | inactive"
        timestamp created_at
        timestamp updated_at
    }

    pegawais {
        bigint id PK
        varchar nama
        varchar nik UK "unique"
        varchar jenis_kelamin "nullable"
        varchar foto "nullable, path file"
        varchar pendidikan
        bigint satker_id FK "constrained"
        enum status "aktif | non_aktif"
        varchar file_ktp "nullable, path file"
        varchar file_kk "nullable, path file"
        timestamp created_at
        timestamp updated_at
        timestamp deleted_at "SoftDeletes"
    }

    pegawai_requests {
        bigint id PK
```

```
        bigint pegawai_id FK "nullable, nullOnDelete"
        bigint satker_id FK "constrained"
        bigint requested_by FK "constrained to users"
        enum action_type "create | update | delete"
        json data_payload "nullable, snapshot data"
        enum status "pending | approved | rejected"
        bigint approved_by FK "nullable, nullOnDelete to users"
        timestamp approved_at "nullable"
        timestamp created_at
        timestamp updated_at
    }

    sessions {
        varchar id PK
        bigint user_id FK "nullable"
        varchar ip_address "nullable"
        text user_agent "nullable"
        longtext payload
        int last_activity
    }

    password_reset_tokens {
        varchar email PK
        varchar token
        timestamp created_at "nullable"
    }

    satkers ||--o{ users : "1 Satker → banyak User"
    satkers ||--o{ pegawais : "1 Satker → banyak Pegawai"
    satkers ||--o{ pegawai_requests : "1 Satker → banyak Request"
    users ||--o{ pegawai_requests : "requested_by"
    users ||--o{ pegawai_requests : "approved_by"
    pegawais ||--o{ pegawai_requests : "pegawai_id"
    users ||--o{ sessions : "user_id"
```

Penjelasan Relasi Database

Relasi	Tipe	Foreign Key	Constraint
`satkers` → `users`	One-to-Many	`users.satker_id`	`nullOnDelete` (jika satker dihapus, satker_id jadi NULL)
`satkers` → `pegawais`	One-to-Many	`pegawais.satker_id`	`constrained` (tidak bisa hapus satker jika masih ada pegawai)
`satkers` → `pegawai_requests`	One-to-Many	`pegawai_requests.satker_id`	`constrained`

Relasi	Tipe	Foreign Key	Constraint
`users` → `pegawai_requests` (pembuat)	One-to-Many	`pegawai_requests.requested_by`	`constrained`
`users` → `pegawai_requests` (approver)	One-to-Many	`pegawai_requests.approved_by`	`nullOnDelete`
`pegawais` → `pegawai_requests`	One-to-Many	`pegawai_requests.pegawai_id`	`nullable`, `nullOnDelete`

> **PENTING:** Tabel `pegawais` menggunakan **SoftDeletes** — data tidak benar-benar dihapus dari database, hanya ditandai dengan `deleted_at`. Ini memastikan riwayat `pegawai_requests` tetap bisa di-trace.

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2. Flowchart Alur Kerja Aplikasi (System Flow)

2A. Alur Autentikasi & Otorisasi

```
flowchart TD
    START(["🌐 User Akses Aplikasi"]) --> GUEST{"Sudah Login?"}

    GUEST -- Belum --> LOGIN["Halaman Login
POST /login"]
    LOGIN --> VALIDATE{"Username & Password Valid?"}
    VALIDATE -- Tidak --> ERRLOGIN["❌ Tampilkan Error"]
    ERRLOGIN --> LOGIN
    VALIDATE -- Ya --> CHECKSTATUS{"Status Akun Active?"}
    CHECKSTATUS -- Tidak --> ERRINACTIVE["❌ Akun Nonaktif"]
    ERRINACTIVE --> LOGIN
    CHECKSTATUS -- Ya --> CHECKROLE{"Cek Role User"}

    GUEST -- Sudah --> CHECKROLE

    CHECKROLE -- super_admin --> SA_DASH["🏠 Dashboard Super Admin
Full Access"]
    CHECKROLE -- admin_satker --> OP_DASH["🏠 Dashboard Operator
Scoped to Satker"]
```

2B. Alur Utama Super Admin

```
flowchart TD
    SA["👑 Super Admin Dashboard"] --> SA_MENU{"Pilih Menu"}

    SA_MENU --> M_SATKER["📋 Kelola Satker"]
```

```
    SA_MENU --> M_USER["👥 Kelola User"]
    SA_MENU --> M_PEGAWAI["📄 Kelola Pegawai"]
    SA_MENU --> M_APPROVAL["☑ Approval Center"]
    SA_MENU --> M_PROFILE["👤 Edit Profile"]

    %% ── Satker CRUD ──
    M_SATKER --> S_LIST["Lihat Daftar Satker
GET /satker"]
    S_LIST --> S_CREATE["Tambah Satker"]
    S_LIST --> S_EDIT["Edit Satker"]
    S_LIST --> S_DELETE["Hapus Satker"]
    S_CREATE --> S_STORE["POST /satker → DB satkers"]
    S_EDIT --> S_UPDATE["PUT /satker/{id} → DB satkers"]
    S_DELETE --> S_DESTROY["DELETE /satker/{id} → DB satkers"]

    %% ── User Management ──
    M_USER --> U_LIST["Lihat Daftar User
GET /users"]
    U_LIST --> U_CREATE["Tambah User Baru"]
    U_LIST --> U_EDIT["Edit User"]
    U_LIST --> U_DELETE["Hapus User"]
    U_LIST --> U_TOGGLE["Toggle Status Active/Inactive"]
    U_CREATE --> U_STORE["POST /users → DB users"]
    U_EDIT --> U_UPDATE["PUT /users/{id} → DB users"]
    U_DELETE --> U_DESTROY["DELETE /users/{id} → DB users"]
    U_TOGGLE --> U_PATCH["PATCH /users/{id}/toggle-status"]

    %% ── Pegawai Direct CRUD ──
    M_PEGAWAI --> P_LIST["Lihat Daftar Pegawai
GET /pegawai
Filter: satker, search nama/NIK"]
    P_LIST --> P_CREATE["Tambah Pegawai"]
    P_LIST --> P_SHOW["Detail Pegawai"]
    P_LIST --> P_EDIT["Edit Pegawai"]
    P_LIST --> P_DELETE["Hapus Pegawai"]
    P_LIST --> P_EXPORT["📥 Export Excel"]
    P_CREATE --> P_STORE_SA["POST /pegawai
→ INSERT langsung ke DB pegawais"]
    P_EDIT --> P_UPDATE_SA["PUT /pegawai/{id}
→ UPDATE langsung ke DB pegawais"]
    P_DELETE --> P_DESTROY_SA["DELETE /pegawai/{id}
→ Soft Delete di DB pegawais"]

    %% ── Approval Workflow ──
    M_APPROVAL --> A_LIST["Lihat Semua Request
GET /approvals
Filter: status"]
    A_LIST --> A_REVIEW{"Review Request"}
    A_REVIEW -- Setuju --> A_APPROVE["POST /approvals/{id}/approve"]
    A_REVIEW -- Tolak --> A_REJECT["POST /approvals/{id}/reject"]

    A_APPROVE --> A_APPLY{"Terapkan Action"}
    A_APPLY -- create --> A_INSERT["INSERT ke DB pegawais"]
    A_APPLY -- update --> A_UPDATE_PEG["UPDATE di DB pegawais"]
```

```
    A_APPLY -- delete --> A_SOFT_DEL["Soft Delete di DB pegawais"]

    A_REJECT --> A_CLEAN["Hapus file yang sudah diupload
Update status → rejected"]

    style SA fill:#1a1a2e,color:#fff,stroke:#e94560
    style M_APPROVAL fill:#0f3460,color:#fff
    style A_APPROVE fill:#16a085,color:#fff
    style A_REJECT fill:#e74c3c,color:#fff
```

2C. Alur Operator (Admin Satker) — Dengan Approval Workflow

```
flowchart TD
    OP["🔧 Operator Dashboard
Scoped ke Satker sendiri"] --> OP_MENU{"Pilih Menu"}

    OP_MENU --> OP_PEG["📄 Kelola Pegawai
Hanya milik Satker sendiri"]
    OP_MENU --> OP_PROF["👤 Edit Profile"]

    OP_PEG --> OP_LIST["Lihat Daftar Pegawai
Auto-filter by satker_id"]

    OP_LIST --> OP_CREATE["Tambah Pegawai Baru"]
    OP_LIST --> OP_SHOW["Detail Pegawai"]
    OP_LIST --> OP_EDIT["Edit Pegawai"]
    OP_LIST --> OP_DELETE["Hapus Pegawai"]
    OP_LIST --> OP_EXPORT["📥 Export Excel"]

    OP_CREATE --> REQ_C["Buat PegawaiRequest
action_type: create
status: pending"]
    OP_EDIT --> REQ_U["Buat PegawaiRequest
action_type: update
status: pending"]
    OP_DELETE --> REQ_D["Buat PegawaiRequest
action_type: delete
status: pending"]

    REQ_C --> WAIT["⏳ Menunggu Persetujuan
Super Admin"]
    REQ_U --> WAIT
    REQ_D --> WAIT

    WAIT --> SA_REVIEW{"Super Admin Review"}
    SA_REVIEW -- Approved --> APPLIED["✅ Perubahan Diterapkan
ke DB pegawais"]
    SA_REVIEW -- Rejected --> DITOLAK["❌ Permintaan Ditolak
File dicleanup"]

    style OP fill:#2c3e50,color:#fff,stroke:#3498db
```

```
    style WAIT fill:#f39c12,color:#000
    style APPLIED fill:#27ae60,color:#fff
    style DITOLAK fill:#c0392b,color:#fff
```

2D. Alur Data pada Tabel `pegawai_requests` (Approval Lifecycle)

```
stateDiagram-v2
    [*] --> Pending : Operator submit request

    Pending --> Approved : Super Admin approve
    Pending --> Rejected : Super Admin reject

    Approved --> [*] : Action diterapkan ke DB
    Rejected --> [*] : File di-cleanup

    state Pending {
        [*] --> WaitingReview
        WaitingReview : data_payload berisi snapshot data
        WaitingReview : action_type = create/update/delete
        WaitingReview : approved_by = NULL
    }

    state Approved {
        [*] --> Applied
        Applied : approved_by = Super Admin ID
        Applied : approved_at = timestamp
        Applied : Pegawai record diupdate
    }

    state Rejected {
        [*] --> Cleaned
        Cleaned : approved_by = Super Admin ID
        Cleaned : approved_at = timestamp
        Cleaned : File KTP/KK dihapus
    }
```

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3. Ringkasan Arsitektur

Role-Based Access Control (RBAC)

Fitur	Super Admin	Operator (Admin Satker)
Dashboard	☑ Semua data	☑ Data satker sendiri
Kelola Satker	☑ CRUD langsung	✕
Kelola User	☑ CRUD + toggle status	✕
Kelola Pegawai	☑ CRUD langsung	⏳ Via approval request

Fitur	Super Admin	Operator (Admin Satker)
Approval Center	☑ Review & approve/reject	✗
Export Excel	☑	☑
Edit Profile	☑	☑

Data Teraktual dari Database

Berdasarkan JSON ekspor:

Tabel	Jumlah Record	Keterangan
`satkers`	3	Bagian Umum, Bagian Keuangan, Bagian Kepegawaian
`users`	2	1 Super Admin, 1 Operator (Satker Umum)
`pegawais`	20	8 di Satker 1, 6 di Satker 2, 6 di Satker 3
`pegawai_requests`	0	Belum ada request yang diajukan
`sessions`	0	Tidak ada session aktif

> **CATATAN:** Semua 14 migration sudah berhasil dijalankan dalam 1 batch, menandakan database dalam kondisi konsisten dan up-to-date.